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Preface

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e. positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$. The multiplicative group of complex numbers of absolute value 1
ζ_m	A primitive m-th root of unity
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements
$(\mathbb{F}_q^*)^{(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
E, F, L, K	Generic notation for finite fields or finite extensions of finite fields
Characters of Finite Abelian Groups	
G	A finite abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$, a subgroup of $\widehat{G}$
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$

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Symbol	**Meaning**
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma : \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i \,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E by $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ with $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
ψ_j	Multiplicative character defined by $\psi_j(g^k) = e^{2\pi i jk/(q-1)}$, for $0 \le j \le q-2$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = 1$ if α is a square in $\mathbb{F}_q^*$, and -1 otherwise
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E by $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ with $[E : \mathbb{F}_q] = r$
Exponential Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$, for $\lambda_i \in \boldsymbol{\mathcal{M}}_q$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; the special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at 0; the special case $\alpha = 0$ of J_α
Dirichlet L-functions over $\mathbb{F}_q[X]$	
$\zeta(s)$	Riemann-Zeta function for $s \in \mathbb{C}$
$L(s, \chi)$	Dirichlet L-function: $\sum_{h \in \Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z, \chi)$	Power series associated to $L(s, \chi) = \sum_{h \in \Phi} \chi(h)\, z^{\deg(h)}$, satisfying $\overline{L}(q^{-s}, \chi) = L(s, \chi)$
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
L_k	k-th coefficient of $z \frac{d}{dz} \log \overline{L}(z, \chi) = \sum_{h \in \mathrm{Irr}(\Phi),\, \deg(h) \mid k} \deg(h)\, \chi(h)^{k/\deg(h)}$

Contents

Chapter — List of Notations — Page 4

Chapter 1 — Equations over Finite Fields — Page 7

1.1 Some Elementary Results 7
1.1.1 Univariate Polynomials 7
1.1.2 Chevalley–Warning Theorem 8
1.1.3 Lower Bounds on the Number of Solutions 10
1.1.4 Upper Bounds on the Number of Solutions 12
1.1.5 Average Number of Solutions 14
1.2 Character Sums with Polynomials 15
1.3 Special L-functions 15

Chapter 2 — Bibliography — Page 16

Chapter A — Appendix — Page 17

A.1 Meromorphic and Analytic Continuation 17
A.2 Fermat's Little Theorem 17
A.3 Waring's Formula 18

Chapter — Index — Page 19

Chapter 1

Equations over Finite Fields

In this chapter we consider polynomial equations of the form $f_1(X_1, \ldots, X_n) = f_2(X_1, \ldots, X_n)$ with $f_1, f_2 \in \mathbb{F}_q[X_1, \ldots, X_n]$. We study the number of solutions of such equations in $\mathbb{F}_q$. This equation is equivalent to equations of the form $f(X_1, \ldots, X_n) = 0$ where $f = f_1 - f_2$. From now on we use the notation $\mathcal{V}(f)$ to denote the set of solutions of $f(X_1, \ldots, X_n) = 0$ and $Z(f)$ denotes the number of solutions. Hence $Z(f) = |\mathcal{V}(f)|$. Similarly we define $\overline{Z}(f)$ and $\overline{\mathcal{V}}(F)$ to denote set of non-trivial solutions and number of non-trivial solutions of $f(X_1, \ldots, X_n) = 0$ respectively. For a system of polynomial equations $f_1 = 0, \ldots, f_k = 0$ with $f_1, \ldots, f_k \in \mathbb{F}_q[X_1, \ldots, X_n]$ we write $\mathcal{V}(f_1, \ldots, f_k) = \mathcal{V}(f_1) \cap \cdots \cap \mathcal{V}(f_k)$ for the set of common solutions and $Z(f_1, \ldots, f_k) = |\mathcal{V}(f_1, \ldots, f_k)|$ for their count. And similarly we define $\overline{Z}(f_1, \ldots, f_k)$ and $\overline{\mathcal{V}}(f_1, \ldots, f_k)$. This notion is coming the notation used in the discussion of affine varieties in algebraic geometry.

In some cases we can give exact formula for the number of solutions and in general case we can give bounds on the number of solutions. To find bounds on numbers we use multiplicative or additive characters applied on polynomial evaluations to get. So we look into character sums of polynomials too. The problem of evaluating such character sums explicitly is very difficult. That's where we try to get bounds.

1.1 Some Elementary Results

In this section we give some elementary results on the number of solutions. We present some classical theorems such as those of König-Rados, Chevalley, and Warning. We also establish elementary upper bounds for the number of solutions and results on the expected order of magnitude.

1.1.1 Univariate Polynomials

We start of with univariate polynomials. Let $f \in \mathbb{F}_q[X]$ is an univariate polynomial Naturally the number of solutions in $\mathbb{F}_q$ for the equation $f(X) = 0$ is basically the degree of $\gcd(f(X), X^q - X)$. Since checking if 0 is a solution or not is very easy to check we'll only consider non-zero solutions. So the number of non-zero solutions of $f(X) = 0$ is basically the degree of $\gcd(f(X), X^{q-1} - 1)$. Therefore we may always assume without loss of generality that $\deg(f) \leq q - 1$. Suppose

$$f(X) = a_{q-1} \cdot X^{q-1} + a_{q-2} \cdot X^{q-2} + \cdots + a_1 \cdot X + a_0$$

where $a_i \in \mathbb{F}_q$ for all $0 \leq i \leq q - 1$. Since for all $\alpha \in \mathbb{F}_q^*$ we have $\alpha^{q-1} = 1$ the variable X^{q-1} will always evaluate to be 1 for all non-zero values of X. Therefore the number of solutions of f is same as of the equation $a_{q-2} \cdot X^{q-2} + \cdots + a_1 \cdot X + (a_0 + a_{q-1}) = 0$. Therefore we can always assume $\deg(f) \leq q - 2$. So suppose we have a polynomial $f \in \mathbb{F}_q[X]$ such that

$$f(X) = a_{q-2} \cdot X^{q-2} + \cdots + a_1 \cdot X + a_0$$

where $a_i \in \mathbb{F}_q$ for all $0 \leq i \leq q - 2$.

Now we can also analyze the number of solutions of f using matrix methods. From f we can define the following matrix

$$M_f = \begin{bmatrix} a_0 & a_1 & \cdots & a_{q-2} \\ a_1 & a_2 & \cdots & a_0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{q-2} & a_0 & \cdots & a_{q-3} \end{bmatrix}$$

This matrix is a *left circulant matrix* i.e. each row is obtained from the preceding row by a cyclic shift of the entries to the left. The number of non-zero solutions of $f(X) = 0$ and the rank of the matrix M_f are very related.

Theorem 1.1.1 König-Rados Theorem

Let $f(X) = \sum_{i=0}^{q-2} a_i \cdot X^i \in \mathbb{F}_q[X]$ be an univariate polynomial. Then

$$\overline{Z}(f) = q - 1 - \operatorname{rank}(M_f)$$

Proof: Since $\deg(f) \leq q-2$ we have $\overline{Z}(f) \leq q-2$. $\overline{Z}(f) = q-1$ rises only when f is a zero polynomial i.e. M_f is a zero matrix and in that case $\operatorname{rank}(M_f) = 0$ which satisfies the theorem. Let γ be the primitive element of $\mathbb{F}_q$. Then consider the *Vandermonde Matrix*, $V(\gamma, \gamma^2, \ldots, \gamma^{q-1})$:

$$V = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \gamma & \gamma^2 & \cdots & \gamma^{q-1} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma^{q-2} & \gamma^{2(q-2)} & \cdots & \gamma^{(q-1)(q-2)} \end{bmatrix}$$

Then using $\gamma^{i\cdot(q-1)} = 1$ for all $i \in [q-1]$ we obtain

$$M_f \cdot V = \begin{bmatrix} f(\gamma) & f(\gamma^2) & \cdots & f(\gamma^{q-1}) \\ \gamma^{-1} f(\gamma) & \gamma^{-2} f(\gamma^2) & \cdots & \gamma^{-(q-1)} f(\gamma^{q-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma^{-(q-2)} f(\gamma) & \gamma^{-2(q-2)} f(\gamma^2) & \cdots & \gamma^{-(q-1)(q-2)} f(\gamma^{q-1}) \end{bmatrix}$$

Now for any $i \in [q-1]$ we have $f(\gamma^i) = 0$ if and only if the ith row becomes zero. Therefore the $\operatorname{rank}(M_f \cdot V) = q - 1 - \overline{Z}(f)$. Since V is the Vandermonde matrix V and $M_f \cdot V$ have the same rank. So the theorem follows. ■

In the next subsections we will give simple upper and lower bounds for solutions multivariate polynomial equations or system of polynomial equations.

1.1.2 Chevalley–Warning Theorem

We begin with two elementary lemmas about power sums that will serve as the key tools.

Lemma 1.1.2

Let k be a non-negative integer. Suppose for $k = 0$, $0^0 = 1$. Then

$$\sum_{\alpha \in \mathbb{F}_q} \alpha^k = \begin{cases} 0 & \text{if } k = 0 \text{ or } q-1 \nmid k \\ -1 & \text{if } q-1 \mid k \end{cases}$$

Proof: For $k = 0$ the sum becomes $q \equiv 0 \bmod q$. So suppose $k > 0$. Let γ is the primitive element of $\mathbb{F}_q$. Then we have

$$\sum_{\alpha \in \mathbb{F}_q} \alpha^k = \sum_{\alpha \in \mathbb{F}_q^*} \alpha^k = \sum_{j=0}^{q-2} \gamma^{j \cdot k} = \sum_{j=0}^{q-2} (\gamma^k)^j$$

Now if $q - 1 \mid k$ then $\gamma^k = 1$ so the sum evaluates to be -1. If $q - 1 \nmid k$ then $\sum_{j=0}^{q-2} (\gamma^k)^j = \frac{\gamma^{k(q-1)}-1}{\gamma^k - 1} = 0$. Hence we have the lemma. ■

So using this lemma we can prove if we take the sum over all evaluations of a polynomial then the sum evaluates to be zero.

Lemma 1.1.3

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $\deg(f) < n(q-1)$. Then

$$\sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} f(\alpha_1, \ldots, \alpha_k) = 0$$

Proof: It suffices to prove this property for every monomial of f. So suppose $X_1^{k_1} \cdots X_n^{k_n}$ be a monomial of f where $k_1 + \cdots + k_n < n(q-1)$ and $k_i \in \mathbb{Z}_0$ for all $i \in [n]$. Since $k_1 + \cdots + k_n < n(q-1)$ then by Pigeon Hole Principle there exists $j \in [n]$ such that $k_j < q - 1$. Then

$$\sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} \alpha_1^{k_1} \cdots \alpha_n^{k_n} = \prod_{i=1}^{n} \left(\sum_{\alpha_i \in \mathbb{F}_q} \alpha^{k_i} \right)$$

Then for j, by Lemma 1.1.2 $\sum_{\alpha_j \in \mathbb{F}_q} \alpha^{k_j} = 0$. So we have the lemma. ■

The two lemmas above are the principal tools behind the Chevalley–Warning theorem. It asserts that whenever the total degree of a system is small relative to the number of variables, the number of common zeros is divisible by p — forcing, in particular, that a system with one solution must have a second.

Theorem 1.1.4 Chevalley–Warning Theorem

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $\deg(f) < n$. Then the number of solutions of the equation $f(X_1, \ldots, X_n) = 0$ in $\mathbb{F}_q^n$ is divisible by the characteristic p of $\mathbb{F}_q$.

Moreover, if $f(0, \ldots, 0) = 0$ then there exists a nontrivial solution in $\mathcal{V}(f)$ i.e. $\overline{\mathcal{V}}(f) \neq \emptyset$ or $\overline{Z}(f) \geq 1$.

Proof: We first construct a polynomial and then will use Lemma 1.1.3. So take the polynomial $F(X_1, \ldots, X_n) = 1 - f^{q-1}(X_1, \ldots, X_n)$. By construction F always takes vale 0 or 1. Therefore for any $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ we have

$$F(\alpha_1, \ldots, \alpha_n) = 1 \iff f(\alpha_1, \ldots, \alpha_n) = 0$$

So

$$\sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} F(\alpha_1, \ldots, \alpha_k) = Z(f) \bmod p$$

Now by Lemma 1.1.3 we have

$$\sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} F(\alpha_1, \ldots, \alpha_k) = 0$$

Therefore $p \mid Z(f)$. So we have the first part.

Now since $f(0,\dots,0)=0$ we have $Z(f)\geq 1$. Since p is a prime $p\geq 2$. So by first part $p\mid Z(f)$. Hence $Z(f)\geq 2$. So we have the second part too. ■

Now the bound of $\deg(f)<n$ in Chevalley–Warning Theorem is the best possible bound for the theorem to work. We can construct a polynomial in $\mathbb{F}_q[X_1,\dots,X_n]$ of degree n for which the theorem doesn't hold. Let $\{\alpha_1,\dots,\alpha_n\}$ is the basis of $E=\mathbb{F}_{q^n}$ over $\mathbb{F}_q$. Consider the following polynomial

$$f(X_1,\dots,X_n)=\prod_{j=0}^{n-1}\left(\alpha_1^{q^i}X_1+\cdots+\alpha_n^{q^i}X_n\right)=\prod_{j=0}^{n-1}\sum_{j=1}^{n}\alpha_j^{q^i}X_i$$

Since for all $i\in\{0,1,\dots,n-1\}$, $\alpha_j^{q^i}$ are conjugates of α_j for all $j\in[n]$ with respect to $\mathbb{F}_q$, f is a polynomial over $\mathbb{F}_q$. Suppose for any $(\gamma_1,\dots,\gamma_n)\in\mathbb{F}_q^n$ consider the element $\alpha_1\cdot\gamma_1+\cdots+\alpha_n\cdot\gamma_n\in E$. Then we evaluate f on $(\gamma_1,\dots,\gamma_n)$ and we get

$$f(\gamma_1,\dots,\gamma_n)=\prod_{i=0}^{n-1}\sum_{j=1}^{n}\alpha_j^{q^i}\cdot\gamma_j=\prod_{i=0}^{n-1}\left(\sum_{j=1}^{n}\alpha_j\cdot\gamma_j\right)^{q^i}=N_{E/\mathbb{F}_q}(\alpha_1\cdot\gamma_1+\cdots+\alpha_n\cdot\gamma_n)$$

Hence $f(\gamma_1,\dots,\gamma)=0$ if and only if $N_{E/\mathbb{F}_q}(\alpha_1\cdot\gamma_1+\cdots+\alpha_n\cdot\gamma_n)=0$ which holds only for $\alpha_1\cdot\gamma_1+\cdots+\alpha_n\cdot\gamma_n=0$. Since $\{\alpha_1,\dots,\alpha_n\}$ forms a basis of E over $\mathbb{F}_q$ we have $f(\gamma_1,\dots,\gamma)=0$ if and only if $\gamma_j=0$ for all $j\in[n]$. Hence $Z(f)=1$. The Chevalley–Warning Theorem clearly doesn't hold in this case.

Now we can extend the Chevalley–Warning Theorem to a system of polynomial equations $f_1=0,\dots,f_k=0$ with $f_1,\dots,f_k\in\mathbb{F}_q[X_1,\dots,X_n]$. In this case we are interested in the number of common solutions.

Theorem 1.1.5

Let $f_1,\dots,f_k\in\mathbb{F}_q[X_1,\dots,X_n]$ with $\deg(f_1\cdots f_k)=\deg(f_1)+\cdots+\deg(f_k)<n$. Then $p\mid Z(f_1,\dots,f_k)$ where $p=\mathrm{char}(\mathbb{F}_q)$.

Moreover, if $(0,\dots,0)\in\mathcal{V}(f_1,\dots,f_k)$ then $\overline{\mathcal{V}}(f_1,\dots,f_k)\neq\emptyset$ i.e. $\overline{Z}(f_1,\dots,f_k)\geq 1$.

Proof: Again we construct a polynomial. Consider the polynomial $F\in\mathbb{F}_q[X_1,\dots,X_n]$ such that $F=(1-f_1^{q_1})\cdots(1-f_k^{q-1})$. Therefore for any $\alpha_1,\dots,\alpha_n\in\mathbb{F}_q$ we have $F(\alpha_1,\dots,\alpha_n)=1$ if and only if $f_i(\alpha_1,\dots,\alpha_n)=0$ for all $i\in[k]$. Now $\deg(F)\leq\deg(f_1)+\cdots+\deg(f_k)<n(q-1)$. Now by applying Chevalley–Warning Theorem for F we get the theorem. ■

1.1.3 Lower Bounds on the Number of Solutions

In the next theorem we will show if we have a system of polynomial equations $f_1=0,\dots,f_k=0$ with $f_1,\dots,f_k\in\mathbb{F}_q[X_1,\dots,X_n]$ then for any affine subspace W all its parallel shifts have same number of solutions modulo the characteristic of $\mathbb{F}_q$. If W_1 and W_2 are two affine subspaces of $\mathbb{F}_q^n$ of same dimension then we call they are parallel affine subspaces if they are obtained by translation from the same linear subspace.

Lemma 1.1.6

Let $f_1,\dots,f_k\in\mathbb{F}_q[X_1,\dots,X_n]$. If W_1 and W_2 are two parallel affine subspaces of $\mathbb{F}_q^n$ of dimension d where $d=\deg(f_1)+\cdots+\deg(f_k)$ then

$$|W_1\cap\mathcal{V}(f_1,\dots,f_k)|\equiv|W_2\cap\mathcal{V}(f_1,\dots,f_k)|\bmod p$$

where $p=\mathrm{char}(\mathbb{F}_q)$.

Proof: Now if $W_1 = W_2$, we get $|W_1 \cap \mathcal{V}(f_1, \ldots, f_k)| = |W_2 \cap \mathcal{V}(f_1, \ldots, f_k)|$. So we get the theorem. So we can assume $W_1 \neq W_2$. Now by change of coordinates we can take

$$W_1 = \{(\alpha_1, \ldots, \alpha_n) \in \mathbb{F}_q^n \colon \alpha_i = 0,\ \forall\, i \in [n]\}$$
$$W_2 = \{(\alpha_1, \ldots, \alpha_n) \in \mathbb{F}_q^n \colon \alpha_1 = 1 \text{ and } \alpha_i = 0,\ \forall\, 2 \leq i \leq n\}$$

Again we construct a polynomial $G \in \mathbb{F}_q[X_1, \ldots, X_n]$ where

$$G(X_1, \ldots, X_n) = (-1)^{n-d}(X_1^{q-2} + \cdots X_1 + 1)(X_2^{q-1} - 1) \cdots (X_n^{q-1} - 1)$$

So $\deg(G) = (n-d)(q-1) - 1$.

Observation 1.1. *With the above construction of G for any $(\alpha_1, \ldots, \alpha_n) \in \mathbb{F}_q^n$*

(i) if $(\alpha_1, \ldots, \alpha_n) \in W_1$ then $G(\alpha_1, \ldots, \alpha_n) = -1$

(ii) if $(\alpha_1, \ldots, \alpha_n) \in W_2$ then $G(\alpha_1, \ldots, \alpha_n) = 1$

(iii) if $(\alpha_1, \ldots, \alpha_n) \notin W_1 \cup W_2$ then $G(\alpha_1, \ldots, \alpha_n) = 0$

Then we construct the following polynomial $F \in \mathbb{F}_q[X_1, \ldots, X_n]$ such that

$$F = (1 - f_1^{q-1}) \cdots (1 - f_k^{q-1})G$$

So now we have $\deg(F) \leq d(q-1) + (n-d)(q-1) - 1 = n(q-1) - 1$. Therefore for any $(\alpha_1, \ldots, \alpha_n) \in \mathbb{F}_q^n$, if $(\alpha_1, \ldots, \alpha_n) \in W_1 \cap \mathcal{V}(f_1, \ldots, f_k)$ then $F(\alpha_1, \ldots, \alpha_n) = 1$. If $(\alpha_1, \ldots, \alpha_n) \in W_2 \cap \mathcal{V}(f_1, \ldots, f_k)$ then $F(\alpha_1, \ldots, \alpha_n) = -1$ and 0 elsewhere. So

$$\sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} F(\alpha_1, \ldots, \alpha_n) = |W_1 \cap \mathcal{V}(f_1, \ldots, f_k)| - |W_2 \cap \mathcal{V}(f_1, \ldots, f_k)|$$

Now by using Chevalley–Warning Theorem on F we have the lemma. ■

The Lemma 1.1.6 above can be used to obtain a lower bound on $Z(f_1, \ldots, f_k)$: the Chevalley–Warning Theorem guarantees that $p \mid Z(f_1, \ldots, f_k)$ whenever $d < n$, but by counting how many parallel affine hyperplanes each must contain a solution we can say much more.

Theorem 1.1.7

Let $f_1, \ldots, f_k \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $d = \deg(f_1) + \cdots + \deg(f_k) < n$. If $Z(f_1, \ldots, f_k) \geq 1$ then $Z(f_1, \ldots, f_k) \geq q^{n-d}$

Proof: We will prove this by analyzing two cases.

Case I: Let there exists an affine subspace W_1 of $\mathbb{F}_q^n$ with dimension d such that $|W_1 \cap \mathcal{V}(f_1, \ldots, f_k)| \not\equiv 0 \bmod p$. Then by Lemma 1.1.6 for all parallel affine spaces W_2 of W_1 we have $|W_2 \cap \mathcal{V}(f_1, \ldots, f_k)| \not\equiv 0 \bmod p$. In particular $|W_2 \cap \mathcal{V}(f_1, \ldots, f_k)| \geq 1$. Since W_1 is of dimension d there are q^{n-d} distinct affine subspaces parallel to W_1. Therefore

$$\mathcal{V}(f_1, \ldots, f_k) = \bigsqcup_{\substack{W\colon \text{affine subspace} \\ \text{parallel to } W_1}} \Big(W \cap \mathcal{V}(f_1, \ldots, f_k)\Big)$$

For each such W we have $|W \cap \mathcal{V}(f_1, \ldots, f_k)| \geq 1$. Therefore

$$Z(f_1, \ldots, f_k) = \sum_{\substack{W\colon \text{affine subspace} \\ \text{parallel to } W_1}} |W \cap \mathcal{V}(f_1, \ldots, f_k)| \geq q^{n-d}$$

Therefore we have the theorem in this case.

Case II: We have $|W \cap \mathcal{V}(f_1, \ldots, f_k)| \equiv 0 \bmod p$. for all affine subspaces W of $\mathbb{F}_q^n$ of dimension d. Since $Z(f_1, \ldots, f_k) \geq 1$ there exists $k \in [d]$ such that for any affine subspace W' of dimension k, $|W' \cap \mathcal{V}(f_1, \ldots, f_k)| \equiv 0 \bmod p$ but there is an affine subspace W'' of dimension $(k-1)$ such that $|W'' \cap \mathcal{V}(f_1, \ldots, f_k)| \not\equiv 0 \bmod p$. Now consider all affine subspaces W' of dimension k containing W''. There are exactly $\frac{q^{n-(k-1)}-1}{q-1}$ many such affine subspaces. For each such affine subspace W' we have

$$|(W' \setminus W'') \cap \mathcal{V}(f_1, \ldots, f_k)| = |W' \cap \mathcal{V}(f_1, \ldots, f_k)| - |W'' \cap \mathcal{V}(f_1, \ldots, f_k)| \not\equiv 0 \bmod p$$

So $|(W' \setminus W'') \cap \mathcal{V}(f_1, \ldots, f_k)| \geq 1$. Therefore we have

$$\mathcal{V}(f_1, \ldots, f_k) = \Big(W'' \cap \mathcal{V}(f_1, \ldots, f_k)\Big) \sqcup \bigsqcup_{\substack{W' \text{: affine subspace} \\ \text{of dimension } k \\ \text{contains } W''}} \Big((W' \setminus W'') \cap \mathcal{V}(f_1, \ldots, f_k)\Big)$$

Hence we have

$$Z(f_1, \ldots, f_k) = |W'' \cap \mathcal{V}(f_1, \ldots, f_k)| + \sum_{\substack{W' \text{: affine subspace} \\ \text{of dimension } k \\ \text{contains } W''}} |(W' \setminus W'') \cap \mathcal{V}(f_1, \ldots, f_k)| \geq 1 + \frac{q^{n-(k-1)}-1}{q-1} > q^{n-d}$$

So we have the theorem. ■

The lower bound on the number of solutions above is best possible even for $k = 1$ i.e. for any positive integers d and n with $d < n$ there is a polynomial $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ of degree d such that $f(X_1, \ldots, X_n) = 0$ has exactly q^{n-d} solutions in $\mathbb{F}_q^n$. Consider the polynomial defined in the support Chevalley–Warning Theorem, $\prod_{i=0}^{n-1} \sum_{j=1}^{n} \alpha_j^{q^i} X_j$ where $\{\alpha_j : j \in [n]\}$ was a basis of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$. But in this example we take the same polynomial but on d variables but the solution space is over $\mathbb{F}_{q^n}$. Hence our polynomial in this case is $f(X_1, \ldots, X_n) = g(X_1, \ldots, X_d) = \prod_{i=0}^{n-1} \sum_{j=1}^{d} \alpha_j^{q^i} X_j$. Since f had only one solution, g has a solution at $(\gamma_1, \ldots, \gamma_n)$ if $\gamma_i = 0$ for all $i \in [d]$. Therefore g has exactly q^{n-d} solutions in $\mathbb{F}_q^n$.

1.1.4 Upper Bounds on the Number of Solutions

We now give upper bounds on the number of solutions of a polynomial equation. The first is the celebrated Schwartz–Zippel lemma, which bounds the probability of a random point being a root in terms of total degree.

Lemma 1.1.8 Schwartz-Zippel Lemma

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $\deg(f) = d$ be a non-zero polynomial. If $S \subseteq \mathbb{F}_q$ be any subset then

$$\mathop{\mathbb{P}}_{\alpha_1, \ldots, \alpha_n \in S}[f(\alpha_1, \ldots, \alpha_n) = 0] \leq \frac{d}{|S|}$$

This lemma immediately gives an upper bound of $d \cdot q^{n-1}$ solutions for the equations $f(X_1, \ldots, X_n) = 0$ in $\mathbb{F}_q^n$. Hence

Corollary 1.1.9

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $\deg(f) = d$ be a non-zero polynomial. Then the equation $f(X_1, \ldots, X_n) = 0$ has at most $d \cdot q^{n-1}$ solutions in $\mathbb{F}_q^n$.

We can give another probabilistic upper bound when we control the degree of f in each variable separately rather than just the total degree.

Theorem 1.1.10 Individual Degree Bound

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ be a non-zero polynomial with $\deg_{X_i}(f) \leq d$ for all $i \in [n]$. Then for $(\alpha_1, \ldots, \alpha_n)$ chosen uniformly at random from $\mathbb{F}_q^n$,

$$\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_n) = 0] \leq 1 - \left(1 - \frac{d}{q}\right)^n$$

Proof: We proceed by induction on n. For the base case $n = 1$, since $f \in \mathbb{F}_q[X_1]$ is non-zero with $\deg_{X_1}(f) \leq d$, it has at most d roots in $\mathbb{F}_q$. Hence $\mathop{\mathbb{P}}_{x_1 \in \mathbb{F}_q}[f(x_1) = 0] \leq d/q = 1 - (1 - d/q)^1$.

Assume the bound holds for $n - 1$ variables. Let $t = \deg_{X_n}(f)$. Then write f as a polynomial in X_n with coefficients in $\mathbb{F}_q[X_1, \ldots, X_{n-1}]$:

$$f(X_1, \ldots, X_n) = \sum_{i=0}^{t} f_i(X_1, \ldots, X_{n-1}) \cdot X_n^k.$$

Since $\deg_{X_j}(f) \leq d$ for each $j \in [n-1]$, every f_i satisfies $\deg_{X_j}(f_i) \leq d$.

Let $\alpha_1, \ldots, \alpha_{n-1}$ be uniformly random points on $\mathbb{F}_q$ and α_n be uniform on $\mathbb{F}_q$, independently. Then by induction hypothesis

$$\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) = 0] \leq 1 - \left(1 - \frac{d}{q}\right)^{n-1}$$

Now we have

$$\begin{aligned}
\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_n) = 0] &\leq \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) = 0] + \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_{n-1}, \alpha_n) = 0 \wedge f_t(\alpha_1, \ldots, \alpha_{n-1}) \neq 0] \\
&\leq \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) = 0] + \\
&\qquad \mathop{\mathbb{P}}_{\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_{n-1}, \alpha_n) = 0 \mid f_t(\alpha_1, \ldots, \alpha_{n-1}) \neq 0] \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) \neq 0]
\end{aligned}$$

Let $p := \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) = 0]$. For the second term after fixing $\alpha_1, \ldots, \alpha_{n-1}$ we have $\mathop{\mathbb{P}}_{\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_{n-1}, \alpha_n) = 0 \mid f_t(\alpha_1, \ldots, \alpha_{n-1}) \neq 0] \leq \frac{d}{q}$. Now

$$\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) \neq 0] = 1 - \mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_{n-1} \in \mathbb{F}_q}[f_t(\alpha_1, \ldots, \alpha_{n-1}) = 0]$$

Therefore together we get

$$\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_n) = 0] \leq p + (1 - p)\frac{d}{q} = \frac{d}{q} + \left(1 - \frac{d}{q}\right)p$$

By induction hypothesis $p \leq 1 - \left(1 - \frac{d}{q}\right)^{n-1}$. Therefore we have

$$\mathop{\mathbb{P}}_{\alpha_1,\ldots,\alpha_n \in \mathbb{F}_q}[f(\alpha_1, \ldots, \alpha_n) = 0] \leq \frac{d}{q} + \left(1 - \frac{d}{q}\right)\left[1 - \left(1 - \frac{d}{q}\right)^{n-1}\right] = 1 - \left(1 - \frac{d}{q}\right)^n$$

So we have the lemma. ■

We can give another upper bound based on individual degrees of each variables. If degree of each variable is at most d then there can be at most $d(q^{n-1} - 1)$ many non-trivial solutions. For this we consider f to be *homogeneous* i.e. total degree of each monomial is same.

Theorem 1.1.11

Let $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ be a homogeneous with $\deg(f) = d \geq 1$. Then the equation $f(X_1, \ldots, X_n) = 0$ has at most $d(q^{n-1} - 1)$ non-trivial solutions.

Proof: We use double induction on n and d. For the base case $n = 1$ then $f = aX_1^d$ for some $a \neq 0$, so the only root is $X_1 = 0$, giving $\overline{Z}(f) = 0 \leq d(q^0 - 1) = 0$. If $d = 1$ then $f = a_1X_1 + \cdots + a_nX_n$ is a non-zero linear form whose zero set is an $(n-1)$-dimensional hyperplane, so $\overline{Z}(f) = q^{n-1} - 1 = d(q^{n-1} - 1)$.

Suppose $n > 1$, $d > 1$, and the bound holds for non-constant homogeneous polynomials in at most n variables of degree less than d and for those in fewer than n variables of degree at most d. We consider two cases.

Case 1: $X_1 \mid f$. Write $f = X_1 \cdot g$ where $g \in \mathbb{F}_q[X_1, \ldots, X_n]$ is homogeneous of degree $d - 1$. Every nontrivial solution $(\alpha_1, \ldots, \alpha_n) \in \overline{Z}(f)$ with $\alpha_i \in \mathbb{F}_q$ for all $i \in [n]$, satisfies $\alpha_1 = 0$ or $g(\alpha_1, \ldots, \alpha_n) = 0$. Solutions with $\alpha_1 = 0$: any $(0, \alpha_2, \ldots, \alpha_n) \neq 0$ works, giving $q^{n-1} - 1$ solutions. Solutions with $\alpha_1 \neq 0$: these are nontrivial zeros of g, and by the induction hypothesis on d there are at most $(d-1)(q^{n-1} - 1)$. Hence

$$\overline{Z}(f) \leq (q^{n-1} - 1) + (d-1)(q^{n-1} - 1) = d(q^{n-1} - 1).$$

Case 2: $X_1 \nmid f$. Write $f = \sum_{k=0}^{d} f_k(X_2, \ldots, X_n) \cdot X_1^k$ with f_k homogeneous of degree $d - k$. Since $X_1 \nmid f$ we have $f_0 \neq 0$, so for every $\alpha \in \mathbb{F}_q$ such that $f(\alpha, X_2, \ldots, X_n)$ is a non-zero polynomial of degree d in $n - 1$ variables we have by Schwartz–Zippel Lemma it has at most $d \cdot q^{n-2}$ zeros in $\mathbb{F}_q^{n-1}$. Now summing over $\alpha \in \mathbb{F}_q^*$ gives at most $(q-1) \cdot d \cdot q^{n-2}$ nontrivial solutions with $\alpha_1 \neq 0$. For solutions with $\alpha_1 = 0$: $f(0, X_2, \ldots, X_n) = f_0(X_2, \ldots, X_n)$ is a nonzero homogeneous polynomial in $n - 1$ variables of degree d, so by the induction hypothesis on n it has at most $d(q^{n-2} - 1)$ nontrivial zeros. Altogether, $\overline{Z}(f) \leq (q-1)d \cdot q^{n-2} + d(q^{n-2} - 1) = d\left(q^{n-1} - q^{n-2} + q^{n-2} - 1\right) = d(q^{n-1} - 1)$. ■

1.1.5 Average Number of Solutions

We now investigate the *average number of solutions* of a polynomial equations. Let $d \in \mathbb{N}$. Then define $\Omega_d := \mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$ and $\omega(d)$ be the set of n–tuples $(i_1, \ldots, i_n) \in \mathbb{Z}_0$ such that $i_1 + \cdots + i_n \leq d$. Then

Observation 1.2. *With the definitions of Ω_d and $\omega(d)$ as above we have $|\Omega_d| = q^{|\omega(d)|}$.*

Lemma 1.1.12

With the notations of Ω_d and $\omega(d)$ as defined above we have

$$\frac{1}{|\Omega_d|} \sum_{f \in \Omega_d} Z(f) = q^{n-1}$$

Proof: We have

$$\sum_{f \in \Omega_d} Z(f) = \sum_{f \in \Omega_d} \sum_{\substack{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q \\ f(\alpha_1, \ldots, \alpha_n) = 0}} 1 = \sum_{\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q} \sum_{\substack{f \in \Omega_d \\ f(\alpha_1, \ldots, \alpha_n) = 0}} 1$$

Let for any $f \in \Omega_d$

$$f(X_1, \ldots, X_n) = \sum_{(i_1, \ldots, i_n) \in \omega(d)} f_{i_1, \ldots, i_n} X_1^{i_1} \cdots X_n^{i_n} \quad f_{i_1, \ldots, i_n} \in \mathbb{F}_q, \ \forall\ (i_1, \ldots, i_n) \in \omega(d)$$

Then for a fixed $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ we get to choose the coefficients $f_{i_1, \ldots, i_n}$ for all $(i_1, \ldots, i_n) \in \omega(d) \setminus \{(0, \ldots, 0)\}$ arbitrarily and then determining the constant term $f_{0, \ldots, o}$ to make $f(\alpha_1, \ldots, \alpha_n) = 0$. Thus the number of $f \in \omega_d$ with $f(\alpha_1, \ldots, \alpha_n) = 0$

is equal to $q^{|\omega(d)|-1}$. Therefore $\sum_{f\in\Omega_d} Z(f) = q^n \cdot q^{|\omega(d)|-1} = |\Omega_d| \cdot q^{n-1}$. So we have the lemma. ■

The proof of the above lemma gives us the following observation:

Observation 1.3. *For any $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$, the number of polynomials $f \in \Omega_d$ such that $f(\alpha_1, \ldots, \alpha_n) = 0$ is $q^{|\omega(d)|-1}$.*

The above lemma implies that a polynomial equation in n variables has on average q^{n-1} solutions in $\mathbb{F}_q^n$. So we now consider average deviation from the average number of solutions, q^{n-1}.

Theorem 1.1.13

With the notations of Ω_d and $\omega(d)$ as defined above we have

$$\frac{1}{|\Omega_d|}\sum_{f\in\Omega_d}\left(Z(f) - q^{n-1}\right)^2 = q^{n-1} - q^{n-2}$$

Proof: Let $\alpha = (\alpha_1, \ldots, \alpha_n), \beta = (\beta_1, \ldots, \beta_n) \in \mathbb{F}_q^n$ be any two points. Then we get

$$\sum_{f\in\Omega_d} Z(f)^2 = \sum_{f\in\Omega_d}\left(\sum_{\substack{\alpha\in\mathbb{F}_q^n\\ f(\alpha)=0}} 1\right)^2 = \sum_{f\in\Omega_d}\sum_{\substack{\alpha\in\mathbb{F}_q^n\\ f(\alpha)=0}}\sum_{\substack{\beta\in\mathbb{F}_q^n\\ f(\beta)=0}} 1 = \sum_{\alpha,\beta\in\mathbb{F}_q^n}\sum_{\substack{f\in\Omega_d\\ f(\alpha)=f(\beta)=0}} 1$$

Now if $\alpha = \beta$ then by Observation 1.3 we have $\sum_{f\in\Omega_d,\, f(\alpha)=0} 1 = q^{|\omega(d)|-1}$. Now suppose $\alpha \neq \beta$. This gives two linear equations for the coefficients of f. Therefore there are $q^{|\omega(d)|-2}$ polynomials $f \in \Omega_d$ such that $f(\alpha) = f(\beta) = 0$. Hence we obtain

$$\begin{aligned}\sum_{f\in\Omega_d} Z(f)^2 = &= \sum_{\alpha\in\mathbb{F}_q^n} |\Omega_d| \cdot q^{-1} + \sum_{\alpha,\beta\in\mathbb{F}_q^n,\alpha\neq\beta} |\Omega_d| \cdot q^{-2} && \text{[By } \textit{Observation } 1.2\text{]}\\ &= q^n \cdot |\Omega_d| \cdot q^{-1} + q^n(q^n-1)|\Omega_d| \cdot q^{-2}\\ &= |\Omega_d|\left(q^{n-1} + q^{2n-2} - q^{n-2}\right)\end{aligned}$$

So now we get

$$\begin{aligned}\sum_{f\in\Omega_d}(Z(f) - q^{n-1})^2 &= \sum_{f\in\Omega_d} Z(f)^2 + 2\cdot q^{n-1}\sum_{f\in\Omega_d} Z(f) + q^{2n-2}\sum_{f\in\Omega_d} 1\\ &= |\Omega_d|\left(q^{2n-2} + q^{n-1} - q^{n-2}\right) - 2\cdot q^{n-1}\cdot|\Omega_d|\cdot q^{n-1} + q^{2n-2}\cdot|\Omega_d|\\ &= |\Omega_d|(q^{n-1} - q^{n-2})\end{aligned}$$

So we have the result. ■

From this result we can expect that $|Z(f) - q^{n-1}|$ is often $O\left(q^{(q-1)/2}\right)$. Later we will see instances of such expected behavior for various polynomial equations. In the next sections we will talk about character sums over polynomial evaluations.

1.2 Character Sums with Polynomials

1.3 Special L-functions

We write alternate proof of Davenport-Hasse Theroem from Schmidt 2.6,7,8,9,10

Chapter 2

Bibliography

Chapter A

Appendix

A.1 Meromorphic and Analytic Continuation

The analytic continuation of the Riemann zeta function, and more generally of Dirichlet L-functions, requires the notion of a meromorphic function. We record the relevant definitions and a motivating example here.

Definition A.1.1: Zero, Pole, Order, Residue

Assume f has a convergent Laurent series expansion about z_0:

$$f(z) = a_n(z-z_0)^n + a_{n+1}(z-z_0)^{n+1} + \cdots = \sum_{m=n}^{\infty} a_m(z-z_0)^m,$$

with $a_n \neq 0$. If $n > 0$ we say f has a zero of order n *at z_0; if $n < 0$ we say f has a* pole of order $-n$ *at z_0. If $n = -1$ we say f has a* simple pole with residue a_{-1}. *We denote the order of f at z_0 by* $\operatorname{ord}_f(z_0) = n$.

Definition A.1.2: Meromorphic Function

We say f is meromorphic at z_0 *if there exists a disk about z_0 of radius r and an integer n_0 such that for all z with $|z-z_0| < r$,*

$$f(z) = \sum_{n \geq n_0} a_n(z-z_0)^n.$$

If f is meromorphic at each point in a region, we say f is meromorphic *on that region. If $n_0 \geq 0$, we say f is* ***analytic****.*

A canonical example illustrates the notion of continuation. Consider the geometric series $G(r) = \sum_{n=1}^{\infty} r^n$, which converges for $|r| < 1$ and equals $\frac{1}{1-r}$ there. Define $H(r) = \frac{1}{1-r}$; then H is well defined for all $r \neq 1$ and agrees with G wherever G converges. Since H is defined on a strictly larger domain and has a simple pole at $r = 1$ (with residue -1), we call H a **meromorphic continuation** of G. Had H no poles, we would call it an **analytic continuation**. A function that is analytic at every point of $\mathbb{C}$ — equivalently, one whose Taylor expansion converges everywhere, so that it has no poles anywhere — is said to be **entire**.

A.2 Fermat's Little Theorem

Fermat's Little Theorem is one of the oldest results in elementary number theory, recorded in a 1640 letter from Pierre de Fermat to Frénicle de Bessy. It underlies essentially every computation we perform in the multiplicative group $\mathbb{F}_p^*$

– from the very existence of the canonical additive character of $\mathbb{F}_q$ (where $q = p^r$), through the character congruence $\psi(l) \equiv l^f \pmod{P}$ used in the arithmetic proof of the Davenport–Hasse relation, down to the mod-q reduction of Gauss sums appearing in the proof of the Law of Quadratic Reciprocity. We record it here for completeness.

Theorem A.2.1 Fermat's Little Theorem

Let p be a prime. Then for every integer a we have

$$a^p \equiv a \pmod{p},$$

and in particular, if $p \nmid a$ then $a^{p-1} \equiv 1 \pmod{p}$.

Proof: The non-zero residues modulo p form a group $\mathbb{F}_p^* = \{1, 2, \ldots, p-1\}$ under multiplication of order $p-1$. By Lagrange's theorem every element of a finite group satisfies $g^{|G|} = e$, so $a^{p-1} \equiv 1 \pmod{p}$ for every $a \in \mathbb{F}_p^*$, i.e. every a with $p \nmid a$. Multiplying both sides by a gives $a^p \equiv a \pmod{p}$ for such a; and the congruence $a^p \equiv a \pmod{p}$ is trivially true when $p \mid a$, so it holds for every integer a. ■

A.3 Waring's Formula

Waring's Formula expresses each power-sum symmetric polynomial $s_k = x_1^k + x_2^k + \cdots + x_n^k$ as an explicit polynomial in the elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$. It is the source of the closed-form reduction of lifted Kloosterman sums $K(\chi^{(s)}; a, b)$ to the ground-field sum $K(\chi; a, b)$ in Theorem **??**: applying Waring's formula to $\omega_1^s + \omega_2^s$ with $\sigma_1 = \omega_1 + \omega_2 = -K(\chi; a, b)$ and $\sigma_2 = \omega_1\omega_2 = q$ produces the binomial-coefficient expansion. We state the general identity here for reference.

Theorem A.3.1 Waring's Formula

Let $x_1, \ldots, x_n$ be indeterminates, let

$$\sigma_j = \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} x_{i_1} x_{i_2} \cdots x_{i_j} \qquad (j = 1, \ldots, n)$$

be the elementary symmetric polynomials in $x_1, \ldots, x_n$, and let

$$s_k = x_1^k + x_2^k + \cdots + x_n^k$$

be the kth power-sum. Then for every integer $k \ge 1$,

$$s_k = \sum (-1)^{i_2+i_4+i_6+\cdots} \cdot \frac{(i_1 + i_2 + \cdots + i_n - 1)! \cdot k}{i_1! \cdot i_2! \cdots i_n!} \cdot \sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n},$$

where the summation runs over all n-tuples $(i_1, \ldots, i_n)$ of non-negative integers satisfying $i_1 + 2i_2 + \cdots + ni_n = k$. Each coefficient of $\sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n}$ is an integer.

Index

analytic continuation, 17
analytic function, 17

Chevalley-Warning theorem, 9

entire function, 17

Fermat's little theorem, 18

meromorphic continuation, 17
meromorphic function, 17

pole, 17

residue, 17

Waring's formula, 18

zero
    of a function, 17